

Supplementary Material S3: Data Extraction Schema — Full Column Definitions

Each of the 7 included studies was coded against a 43-column schema; the 42 analytically relevant columns are described below. (The schema's remaining column is an internal reference-manager lookup key with no analytic content and is not reported here.) Columns are grouped by theme; where a group maps to one of this review's three research questions, that mapping is noted in the group heading.

Identity

Column	Type	Description
`PaperID`	string	Short identifier: `LASTNAME_YEAR` (first author only).
`Title`	string	Full paper title, copied verbatim.
`Year`	integer	Year of publication.

Clinical Domain

Column	Type	Controlled vocabulary
`Domain`	categorical	`Radiology` / `EHR` / `Pathology` / `ICU` / `Pharmacy` / `Oncology` / `Cardiology` / `Other`

Coded as the primary clinical domain the AI system targets; `EHR` covers general inpatient/outpatient prediction using electronic health record data where no more specific specialty label applies.

XAI Method and Scope

Column	Type	Controlled vocabulary
`XAI_Method`	categorical	`SHAP` / `LIME` / `GradCAM` / `Attention` / `ANCHOR` / `Counterfactual` / `Prototype` / `RuleExtraction` / `Other` / `Multiple`
`XAI_Scope`	categorical	`Local` / `Global` / `Both`

`XAI\_Scope` is coded as the method was actually used in the paper, not as the method is theoretically capable of. Attention weights are not coded as demonstrating a faithful explanation without additional evaluation evidence.

Study Design and Participants

Column	Type	Controlled vocabulary / Notes
`Study_Design`	categorical	`RCT` / `Observational` / `UserStudy` / `Simulation` / `MethodPaper`
`N_Participants`	integer	Number of human participants in the XAI evaluation specifically (not the model's training sample size); `0` if proxy metrics only.
`Participant_Type`	categorical	`None` / `LayPerson` / `ClinicalTrainee` / `Clinician` / `MixedClinical`

`MethodPaper` denotes a paper whose stated primary contribution is the explanation method itself, using a clinical domain only as a demonstration case, with no systematic evaluation of clinical utility (see Supplementary Material S2, Criterion 3 exception).

Evaluation Type (RQ2)

Column	Type	Values
`Eval_Type`	categorical, multi-code	`ProxyMetric` / `ForwardSim` / `BackwardSim` / `TrustQuestionnaire` /

		`DecisionQuality` / `DownstreamOutcome` / `None`
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All evaluation types present in a paper are coded, semicolon-separated, ordered from lowest to highest ecological validity (e.g., `ProxyMetric; TrustQuestionnaire; DecisionQuality`). `None` is entered only when the paper presents an XAI system with no evaluation component at all. Definitions of each of the six levels are given in Supplementary Material S2, Criterion 3.

Workflow Realism (RQ2)

Column	Type	Values
`Realism_Level`	ordinal	`0` = No evaluation / `1` = Synthetic / `2` = Simulated / `3` = Ecologically representative / `4` = Deployment-embedded
`Realism_Notes`	string	Rationale for boundary-case coding decisions.

Ecological Validity (RQ2)

Each dimension is scored independently of the others.

Column	Type	Values	Dimension
`EV_Participant`	ordinal 0-3	0=None / 1=NonClinician / 2=ClinOutOfRole / 3=ClinInRole	Participant validity
`EV_Task`	ordinal 0-3	0=NoTask / 1=Artificial / 2=Simplified / 3=FullyRepresentative	Task validity
`EV_Environment`	ordinal 0-3	0=NoInterface / 1=CustomResearch / 2=ClinicalLike / 3=ActualClinical	Environment validity
`EV_Outcome`	ordinal 0-3	0=NoneOrProxy / 1=SelfReport / 2=DecisionQuality /	Outcome validity

		3=PatientOutcomes	
`EV_Notes`	string	Boundary calls and mixed-participant-pool documentation.	

A composite score (sum of all four dimensions, range 0-12) is computed at analysis time rather than stored as its own column, since the four-dimension profile itself is the unit of analysis.

Clinician Involvement

Involvement in evaluating the XAI system and involvement in designing it are recorded separately, since a study can involve clinicians in one role without the other.

Column	Type	Values	Definition
`Clinician_Eval`	boolean	`Yes` / `No`	Was a licensed clinician involved at any stage of evaluating the XAI system?
`Clinician_Role`	categorical, multi-code	`Rater` / `CoDesigner` / `EndUser` / `None`	Role of the clinician in the evaluation; `None` if `Clinician_Eval = No`.
`Clinician_Design`	boolean	`Yes` / `No`	Was a clinician involved in designing or developing the XAI system itself?

Outcomes and Trust

Column	Type	Description
`Outcome_Claimed`	string	What the paper states it demonstrates, quoted or closely paraphrased.
`Outcome_Demonstrated`	string	What is actually measured with a validated instrument, comparison condition, and appropriate sample size; blank if nothing is formally demonstrated.

`Trust_Claim`	categorical	`None` / `Calibrated` / `Plausibility` / `Both`
`Trust_Only`	boolean	`Yes` if a trust questionnaire is the sole evaluation of the XAI component present.

`Trust\_Claim` distinguishes a claim that an explanation helps a user calibrate trust to the system's actual reliability (`Calibrated`) from a claim or demonstration that an explanation is merely found convincing or satisfying, without evidence of calibration (`Plausibility`). This distinction underlies this review's trust-calibration-versus-plausibility construct (see Introduction).

Quality and Risk of Bias

Each dimension is scored independently, adapted from QUADAS-2, RoB 2, and TRIPOD elements. This rubric characterizes methodological execution quality and is conceptually distinct from the ecological-validity dimensions above, which characterize evaluation realism.

Column	Type	Values	Dimension
`QR_Participant`	ordinal 0-2	0=Inappropriate / 1=PartiallyAppropriate / 2=Appropriate	Participant appropriateness
`QR_Task`	ordinal 0-2	0=Misaligned / 1=PartiallyAligned / 2=WellAligned	Task fidelity
`QR_Outcome`	ordinal 0-2	0=InappropriateOrUnvalidated / 1=PartiallyValid / 2=ValidAndAppropriate	Outcome measurement validity
`QR_Faithfulness`	ordinal 0-2	0=None / 1=Indirect / 2=Direct	Explanation faithfulness
`QR_Reporting`	ordinal 0-2	0=MajorGaps / 1=Partial / 2=Complete	Reporting completeness
`QR_Notes`	string	Supporting detail for each dimension's score.	

A composite score (sum of all five dimensions, range 0-10) is computed at analysis time rather than stored as its own column.

Method Justification (RQ1)

Column	Type	Values	Definition
`Method_Justification_Type`	categorical	`Computational` / `Cognitive` / `Workflow` / `Mixed` / `NotReported`	Why the paper says it chose its explanation method over alternatives
`Method_Justification_Notes`	string	—	Quote or close paraphrase of the stated justification

Categories were coded inductively rather than as a closed a priori taxonomy: `Computational` covers justification on grounds such as model-agnosticism or theoretical guarantees; `Cognitive` covers justification on human-comprehensibility grounds; `Workflow` covers justification on deployment/integration grounds; `Mixed` records more than one type explicitly given; `NotReported` records the absence of any stated rationale, without inferring one the paper does not state.

Method-Interface Isolation (RQ1/RQ2)

Column	Type	Values	Definition
`Method_Interface_Isolated`	categorical	`Yes` / `No` / `NotApplicable`	Does the study design allow an observed effect to be attributed to the explanation method itself, separate from how it is displayed?

`Yes` requires a design that isolates method from presentation (e.g., multiple methods shown through an identical interface). `No` denotes a single method shown through a single interface with no comparison condition, leaving any effect on comprehension or trust confounded with presentation choices. `NotApplicable` denotes no human-facing interface exists at all.

Framework Classification (RQ2/RQ3)

Column	Type	Values
`DoshiVelez_Category`	categorical	`FunctionallyGrounded` /

		`HumanGrounded` / `ApplicationGrounded`
`VilonLongo_Category`	categorical	`Objective` / `HumanCentred_Qualitative` / `HumanCentred_Quantitative` / `HumanCentred_Mixed` / `NotEvaluated`

`DoshiVelez\_Category` follows Doshi-Velez and Kim's (2017) three-tier evaluation-grounding framework: `FunctionallyGrounded` denotes no human subjects, interpretability assessed via a formal proxy; `HumanGrounded` denotes human subjects performing a simplified or proxy task rather than the real target decision; `ApplicationGrounded` denotes real end-users performing the real target task in a realistic setting.

`VilonLongo\_Category` follows Vilone and Longo's (2021) primary classification of evaluation approaches as objective/automated versus human-centred, with human-centred evaluations further split into qualitative (open-ended) and quantitative (close-ended) methods. This axis is independent of `DoshiVelez\_Category`: the two frameworks classify different properties of an evaluation (how grounded/realistic it is, versus whether and how it involves humans) and are coded independently from the paper's actual method, not from one another.

Regulatory-Relevant Evidence (RQ3)

Operationalizes what evidence would satisfy interpretability-validation expectations emerging in FDA AI/ML-based Software as a Medical Device guidance and EU AI Act Article 13.

Column	Type	Values	Criterion
`Reg_Comprehension`	boolean	`Yes` / `No`	Clinician comprehension of explanation outputs, demonstrated via a comprehension check or correct-interpretation task
`Reg_TrustCalibration`	boolean	`Yes` / `No`	Trust shown to track actual model reliability across varying performance conditions
`Reg_UncertaintyTransparency`	boolean	`Yes` / `No`	Model uncertainty or failure modes

			surfaced to the clinician at the point of use, not merely discussed as a limitation
`Reg_WorkflowSafety`	boolean	`Yes` / `No`	Safe and effective use demonstrated within a realistic clinical workflow
`Reg_Notes`	string	—	Evidence supporting each `Yes`; for `No`, whether the criterion was addressed at all or attempted but insufficient

All four criteria default to `No` in the absence of explicit, demonstrated evidence; a paper's stated future-work intentions or implications are not sufficient grounds for a `Yes` code.

General Notes and Semantic Tags

Column	Type	Description
`Notes`	string	Free-text documentation of borderline coding decisions and details not captured by other columns.
`Tags`	string, multi-tag	Semicolon-separated semantic tags from a controlled vocabulary, supporting cross-cutting retrieval across papers (e.g., `#trust_calibration;#clinician_study;#deployment`).